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H2**Started on** Thursday, 24 September 2026, 8:29 AM**State** Finished**Completed on** Thursday, 24 September 2026, 8:30 AM**Time taken** 53 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a		g		g		t		a		b
s2		g		x		t		x		a		y
										b		

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s1[1000], s2[1000];
6
7      scanf("%s", s1);
8      scanf("%s", s2);
9
10     int m = strlen(s1);
11     int n = strlen(s2);
12
13     int dp[m + 1][n + 1];
14
15     for (int i = 0; i <= m; i++)
16         dp[i][0] = 0;
17
18     for (int j = 0; j <= n; j++)
19         dp[0][j] = 0;
20
21     for (int i = 1; i <= m; i++) {
22         for (int j = 1; j <= n; j++) {
23             if (s1[i - 1] == s2[j - 1])
24                 dp[i][j] = dp[i - 1][j - 1] + 1;
25             else
26                 dp[i][j] = dp[i - 1][j] > dp[i][j - 1]
27                     ? dp[i - 1][j]
28                     : dp[i][j - 1];
29         }
30     }
31
32     printf("%d", dp[m][n]);
33
34     return 0;
35 }
36

```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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